1 Quiz (5 points)

No negative points. Read the questions carefully. Except when explicitly mentioned, only one answer is true.

1. Think about our Cobb-Douglas production function describing the overall economy. Suppose that hours worked in the economy increase while capital stock, TFP and population stay constant.

   ☐ GDP per capita increases.

2. When using the standard Cobb-Douglas production function to account for GDP growth, which of the following is TRUE? (assuming $\alpha = 1/3$)

   ☐ If labor and capital grows at 3% and productivity is constant, output grows at 3%

3. The shares of Arcelor-Mittal (producing high quality steel for construction, investment goods) on the stock market – because of its business – are going to move in general

   ☐ strongly procyclical because the demand for steel from investment goods production is strongly procyclical.

4. During the entire recession phase of the business cycle the economy can be in a positive output gap (pick the true answer)

   ☐ Yes, if the recession is short-lived and current output does not fall below potential output before the next expansion

5. PMIs are good leading indicators because

   ☐ They aggregate early information about production and the demand for intermediate goods.

6. In general, inflation is

   ☐ procyclical and lagging

7. GDP data are often subject to revisions as new data arrives. Which of the following new data does not lead to any change in the GDP of France? (multiple answers are possible)

   ☐ It is discovered that consumers bought €3 billion more furniture that previously estimated. This furniture was manufactured in Sweden.

8. Indicate how the following transaction changes GDP: Mr. Dupont buys a old apartment valued 300,000 Euros. He pays additionally 10,000 Euros fees to the real estate agent, 8,000 Euros (as fees) to a notary and pays 1,000 euros in fire and theft insurance.

   ☐ Consumption of services increases by 19,000 Euros.

9. Which of the following events will NOT increase the unemployment rate according to the ILO definition?

   ☐ 100 job seekers aged between 50 to 55 years old decided to stop looking for jobs.

10. Suppose the Required Reserve Ratio is 2% (TYPO=: 5%). The money multiplier, assuming all deposits (after required reserves on them are stored at the central bank) would be lent out by the banking system is_____?

    ☐ 20.

	Guns		Roses	
year	prices	quantity	prices	quantity
2016	5	100	10	80
2017	6	120	9	150
2018	5	100	8	200

2 Real GDP and CPI Inflation (3 points)

The beautiful country of Freedonia is specialized in the production of only two goods, guns and roses. The table below reproduces the economic outcomes in the last four years:

Here is the data:

i) Taking the 2017 as a reference year calculate:

a) the growth rate of real GDP between the year 2017 and 2018

SOLUTION: The Real GDP of 2017 taking 2017 as reference year is given by: 6*120+9*150=2070. The Real GDP of 2018 taking 2017 as reference year is given by: 6*100+9*200=2400. Thus, the growth rate is (2400-2070)/2070=15,94%

b) the growth rate of the CPI index between the year 2017 and 2018.

SOLUTION: The CPI of 2017 taking 2017 as reference year is given by: 100*(6*120+9*150)/(6*120+9*150)=100. The CPI of 2018 taking 2017 as reference year is given by:100*(5*120+8*150)/(6*120+9*150)=86,95. Thus, the growth rate (inflation) is -13,05%

c) Re-do a) and b) by taking 2016 as a reference year.

SOLUTION: The Real GDP of 2017 taking 2016 as reference year is given by: 5*120+10*150=2100. The Real GDP of 2018 taking 2016 as reference year is given by: 5*100+10*200=2500. Thus, the growth rate is (2500-2100)/2100=19,04%

The CPI of 2017 taking 2016 as reference year is given by: 100*(6*100+9*80)/(5*100+10*80)=101,53. The CPI of 2018 taking 2016 as reference year is given by:100*(5*100+8*80)/(5*100+10*80)=87,69. Thus, the growth rate (inflation) is (87,69-101,53)/101,53=-13,64%

ii) Freedonia's GDP accounts only for the production of guns and roses. Imagine now that Freedonia's inhabitants never bought a single gun in the whole history of Fredonia. Is there necessarily a contradiction in these two sentences? Explain your answer by referring to one of the three GDP accounting approaches.

SOLUTION: There is no contradiction, this is the situation when all guns produced in Freedonia are sold abroad. Thus, in terms of the spending approach to GDP accounting, we have that consumption in guns is zero but exports account for the whole guns production.

iii) In 2018 an unfortunate armistice prevented the sale of 50 freshly produced guns. Do these guns appear in any category of Freedonia's 2018 national accounts? If yes, say which one and for which value. If no, explain why. In any case, does it imply anything for the national GDP accounts in 2019?

SOLUTION: The unsold guns will account as 50*5=250 in inventories. In fact, unsold finished products are registered in inventories at market price (50). There will be no consequences in 2019, the value of guns produced in 2018 will no enter in the GDP of 2019.

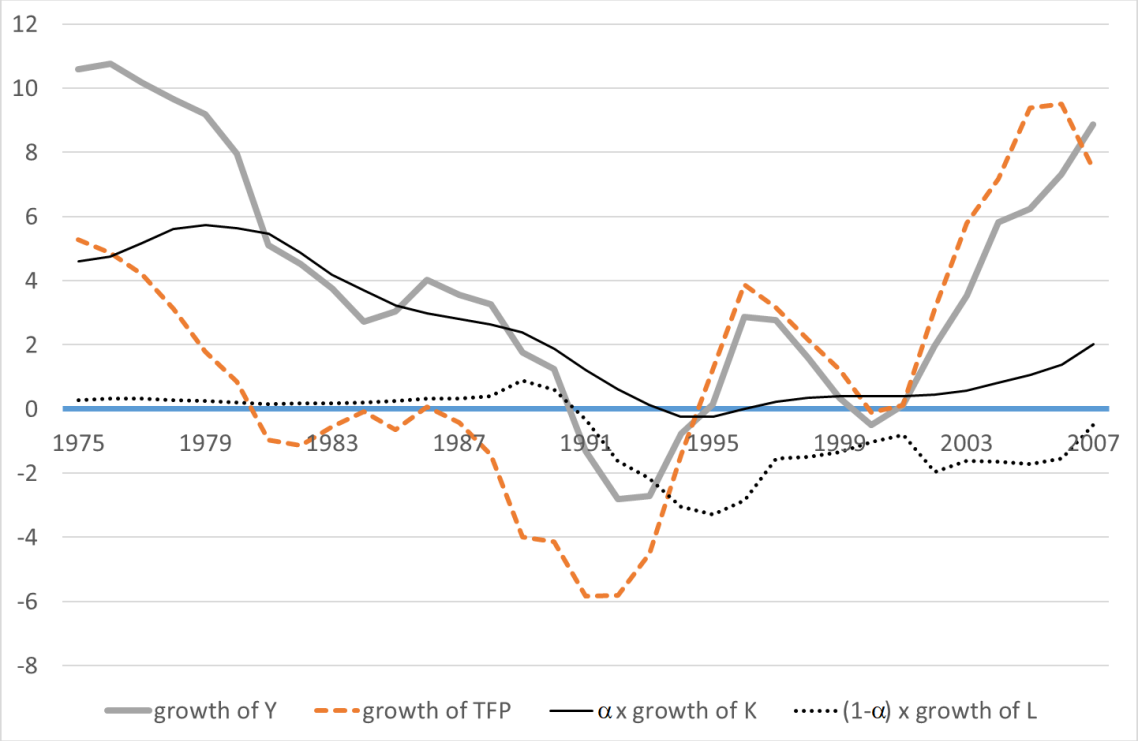


Figure 1: Growth decomposition for Romania over the period 1975-2007 (All quantities are shown in 5-yr averages.)

3 Long run growth and growth accounting: case of Romania. (4 points)

Consider the graph from Figure 1 showing the evolution of GDP growth, TFP, αK and $(1-\alpha)L$ in Romania over the period 1975-2007.

1. Comment on what were the most important drivers of growth of GDP in the period:

   SOLUTION:

   1980-1987: capital accumulation. Changes in TFP or employment were close to zero (though negative for the former and positive for the latter).

   1987-2007: The primary driver of changes in GDP was TFP. Slowing capital accumulation (1987-1991) and then its pickup in the early 2000s was a secondary source. Population growth was a steadily negative factor (approx. 2% a year) from the beginning of 1990s.

2. Comment, citing the data in the graph and arguments from the class, which – total factor productivity growth, constant capital accumulation or increases in hours are necessary to drive growth in the long run.

   SOLUTION: The example of Romania clearly shows that TFP growth is necessary for long-run growth of GDP. In the period between 1980-1995 TFP was declining (esp. quickly in the period 1989-1993) and this lead to a fall in GDP growth from high rates exceeding 5% yearly observed in the 1970s to declines of up to 3% in 1992. The pickup in growth post-1995 is related to fast increases in TFP. Capital accumulation slowed down to zero in 1995 as TFP declined from the beginning of the 1980s.

3. Explain why between 1980 and 1987 capital growth could be gradually slowing when TFP and labor supply (hours worked) were relatively constant (hint: Why could accumulation of capital stop? - recall the arguments from class.).

   SOLUTION: The marginal product of capital falls as more capital is accumulated, holding TFP and labor supply constant. In other words, extra capital (machines) become less productive. Additionally, as TFP would fall (as occurred between 1980-1995) the marginal product would decline with time at any level of capital, and economic agents might have even incentives to disinvest. All these developments would hinder further capital accumulation and cause the growth of capital to fall with time.

4. Fast TFP growth post 1995 is attributed in a large part to foreign direct investment. Explain through what channels foreign firms investing in Romania could contribute in improving the country's TFP.

   SOLUTION: FDI could: allow Romanian firms and plants to adopt new technologies (be related with technological transfers); improve management techniques; increase human capital; increase the access to foreign markets. Perhaps it could also spur the creation of young firms that are especially important for generating TFP.

5. (*harder*) Post 1995 capital accumulation (growth) increases. Why do you think capital accumulation could increase post 1995 for other reasons than caused just by purely capital flows (FDI)?

   SOLUTION: The first answer is that it is related to the capital inflows represented by FDI itself. But, secondly, a strong TFP growth would lead the marginal product of capital to increase as well, and increase the incentives to accumulate it (as it becomes more productive).

6. If population evolved very similarly to the hours worked post 1995, what do you think happened to GDP per capita in the period 1995-2007 given a strong growth of TFP and more capital accumulation? Explain.

   SOLUTION: Hours worked declined in that period. If population evolved in the same way as hours worked, but at the same time aggregate GDP increased, it must be that GDP per capita increased even stronger than GDP itself.

4 Money (3 points)

Anne borrows 1000 euros from Bank A (we assume that Bank A has sufficient excess reserves to grant such a loan) then Anne withdraws 200 euros from her checking account at Bank A to deposit 200 euros on her savings account at Bank B (we assume that savings accounts are also subject to the same reserve requirement as checking accounts).

1. Trace the changes in Bank A's and Bank B's balance sheet arising from the above transactions. (Write down the final changes only, not the intermediate steps). You may assume that the required reserves ratio is 10% on all deposits.

   Bank A's balance sheet (changes only)

Assets	Liabilities
Reserves - 280 (excess) + 80 (required)	checkable deposits + 800
Loans + 1000	other deposits

Bank B's balance sheet (changes only)

Assets	Liabilities
Reserves +180 (excess) + 20 (required) Loans	checkable deposits other deposits +200

2. What are the net changes of M1 and M2 money supply arising from the whole series of transactions? What about M0 ?

   SOLUTION:

   M0 does not change. M1 increases by 800 euros and M2 by 1000 euros.

3. If Bank B did not want to keep excess reserves, what action Bank B is likely to take and what would be the consequences for money supply M1 and M2?

   SOLUTION:

   Bank B would lend money, up to an additional 1800 euros (180/10%). This will increase M1 and M2 if the corresponding deposits are kept as checking accounts and M2 only if the deposits are in form of savings accounts.

4. Suppose Anne buys a Treasury bond worth 100 euros from the central bank with a check drawn on her checking account at Bank B. Trace the changes in Bank's B balance sheet arising from this transaction.

   Bank B's balance sheet (changes only)

Assets		Liabilities
Reserves excess - 90 euros	required - 10 euros	checkable deposits
Loans		other deposits - 100 euros

5. What is the change in M0, M1 and M2 money supply arising from this transaction?

   SOLUTION:

   M0 decreases by 100 euros, so does M2. M1 stays constant.

5 Business cycle (5 points)

Based on the information below, comment on the following points:

- Where is currently the euro area in the cycle?

- In what types of assets (stocks or bonds) should you invest in? In which sectors?

Describe and define carefully each piece of information that you are using.

> .[...] According to Eurostat's second release, euro area real GDP grew by 0.2% on a quarterly basis in the second quarter of 2011
>
> [...] Following six consecutive quarters of positive growth, euro area private consumption declined quarter on quarter by 0.2% in the second quarter of 2011. Consumption is currently 1.2% above its most recent trough in the third quarter of 2009, but 0.5% below its most recent peak in the first quarter of 2008. The weakness in consumption in the second quarter reflects negative growth in retail sales and new car registrations. Of the larger countries, quarterly consumption growth was negative in Germany, France and the Netherlands.
>
> [...] The European Commission's indicator of consumer confidence [...] posted a sharp decline in August and has now reached the same level as in mid-2010.
>
> [...] The output index for the euro area manufacturing sector obtained from the Purchasing Managers' Index (PMI) survey has declined sharply since April, dropping from a level of about 60, which was close to its ten-year high, to somewhat below 50 [...]. Similar developments, albeit slightly less pronounced, are seen in the PMI services business activity index. This index declined from the high level of 57 in March to stand in August only [1.5] points above [...] 50. Developments in other business surveys, such as those of the European Commission, are in line with the results of the PMI survey, and developments at the aggregate euro area level are broadly shared by those in the larger euro area countries.
>
> [...] The improvement in gross fixed capital formation has been weak during the recovery, with the exception of the second quarter of 2010, when investment growth was particularly strong. Investment grew by 0.2% quarter on quarter in the second quarter of 2011, after growing by 1.8% in the first.
>
> [...] Growth in both the residential and non-residential components of construction investment is also likely to have slowed in the second quarter, as suggested by a very modest rise in construction production. Some normalisation was expected following high volatility in previous quarters, which was related to unusual weather conditions in some countries. In addition, the weakness of the construction sector continued in some countries which had seen substantial increases in construction activity before the onset of the recent recession.
>
> [...] After a rebound in euro area trade towards the end of 2010 and during the first quarter of 2011, both quarter-on-quarter import and export growth (in terms of volume) slowed in the second quarter of 2011, to 0.5% and 1.0%, respectively. However, export growth remained above that of imports, leading to a positive net trade contribution to GDP growth. The slowdown in extra-euro area trade volumes was more pronounced than that in intra-euro area trade over the same quarter.
>
> [...] Unemployment rose sharply during the recent recession and for some months thereafter, up from 7.3% at the beginning of 2008 to 10.2% at its peak in mid-2010. Since then, as activity has recovered, the euro area unemployment rate has edged slowly downwards, averaging 10.0% in the first seven months of 2011.

Monthly Bulletin – September 2011 – European Central Bank.

Additional information: in 2011Q2, the contributions to GDP of the different components of demand were:

Elements of solution:
Basic:

GDP	Domestic demand (excluding inventories)	Changes in inventories	Net exports
+ 0.2%	- 0.15%	+0.05%	+0.3%

- Leading procyclical indicators (PMI, imports, ...) are declining.
- Coincident procyclical variables are slowing down or decreasing.
- Unemployment is declining (lagging and counter-cyclical).

This should lead to the conclusion that the economy is entering in a recession. Investors should tilt their portfolios towards investing in bonds. Among stocks, non-durable good producers may perform better.

Importantly: what matters the most is the the precision and the structure of the arguments used by students.

Advanced:

- GDP is increasing despite a fall in the domestic demand. This increase in GDP is in fact due to the improvements in the trade balance, but this improvement stems from a fall in imports that is not compensated by a fall in exports and imports are falling in the beginning of a recession.
- This is the euro-area sovereign debt crisis: so all bonds are not necessarily a good investment opportunity.
- This new recession is not so distant from the previous one: the euro area was just recovering from the previous recession.